INTRODUCTION

Modern portfolio theory makes frequent use of the idea that investment opportunities can be evaluated using expected return as a measure of reward and variance of return as a measure of risk. In Lesson 1, we will develop an understanding of portfolio return and risk metrics. The forecast expected return and variance of return are functions of the returns on the individual portfolio holdings. To begin, the expected return on a portfolio is a weighted average of the expected returns on the securities in the portfolio. When we have estimated the expected returns on the individual securities, we immediately have portfolio expected return. Lesson 2 focuses on forecasting certain portfolio metrics, such as correlations and covariances by looking at the risk and return on the individual components of a portfolio. Lesson 3 introduces various portfolio risk metrics widely used in portfolio management.

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Portfolio Mathematics

Learning Module Overview

- A portfolio's variance measures its expected investment risk and is defined as $\sigma^2(R_p) = E\{[R_p E(R_p)]^2\}$. A portfolio's expected return (E(Rp)) is a weighted average of the expected returns (R_1 to R_n) on the component securities using their respective proportions of the portfolio in currency units as weights (w_1 to w_n):

$$E(R_p) = E(w_1R_1 + w_2R_2 + \ldots + w_nR_n)$$
$$= w_1E(R_1) + w_2E(R_2) + \ldots + w_nE(R_n)$$

- Portfolio variance is affected by both the risk of the individual component assets and their combined risks together as measured by their covariance, which is defined as

$$\sigma^2(R_p) = \sum_{i=1}^{n} \sum_{j=1}^{n} w_i w_j \mathrm{Cov}(R_i, R_j).$$

- Covariance of returns can be negative (an average negative relationship between returns), zero if returns on the assets are unrelated, or positive (an average positive relationship between returns). Correlation, like covariance, measures linear association and ranges between –1 (strongly inverse) to +1 (strongly direct), with 0 indicating no relationship.
- The covariance of portfolio returns can be estimated using a joint probability function of random variables. Defined on variables X and Y, as P(X,Y), which gives the probability of joint occurrences of their values. For example, P(X=3, Y=2), is the probability X equals 3 and Y equals 2.
- A formula for computing the covariance between random variables R_A and R_B, such as the different assets of a portfolio, is

$$\mathrm{Cov}(R_A, R_B) = \sum_i \sum_j P(R_{A,i}, R_{B,j})(R_{A,i} - ER_A)(R_{B,j} - ER_B).$$

  The value is derived by summing all possible deviation cross-products weighted by the appropriate joint probability.
- The joint probability function simplifies for independent variables, defined for two random variables *X* and *Y* if and only if $P(X,Y) = P(X)P(Y)$. The expected value of the product of both independent and uncorrelated random variables is the product of their expected values.
- An application of normal distribution theory to practical investment problems involves safety-first rules. These focus on reducing the shortfall risk, defined as portfolio value (or portfolio return) falling below some minimum acceptable level over some time horizon,
- The safety-first ratio is defined as (SFRatio = $[E(R_P) - R_L]/\sigma_P$, where $E(R_P)$ is expected portfolio return, R_L is a predetermined minimum threshold level for a variable of interest like portfolio return, and σ_P is portfolio standard deviation. When R_L is the risk-free rate, the safety-first ratio is equivalent to the Sharpe ratio.
- Roy's safety-first criterion states that the optimal portfolio minimizes the probability that portfolio return, R_P, will fall below R_L. For a portfolio with a given safety-first ratio (SFratio), the probability that its return will be less than R_L is *Normal*(–SFRatio), and the safety-first optimal portfolio has the lowest such probability. The criterion is implemented by first calculating each potential portfolio's SFRatio and then choosing the portfolio with the highest SFRatio.

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